

1202SCG Calculus 1
1020ENG Engineering Mathematics 2

Week 10 Lecture

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Filling a Tank

A hose fills a tank at a rate of

$R(t)$ litres per minute,

where t is measured in minutes.

The volume added over a time interval depends on both the rate and the length of time for which that rate acts.

How much water enters the tank during the first three minutes?

A Constant Flow Rate

A hose fills a tank at 4 litres/minute for three minutes. How much water enters the tank?

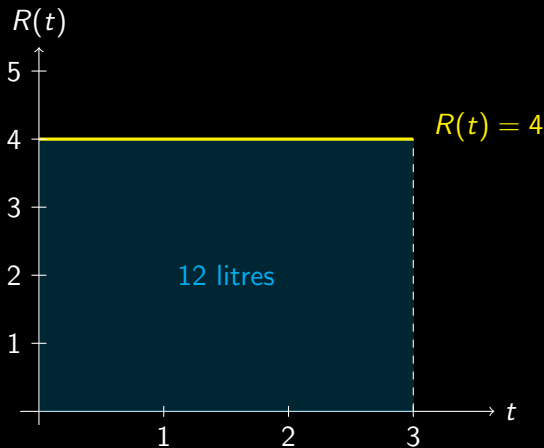
The rate function is

$$R(t) = 4, \quad 0 \leq t \leq 3.$$

Since the rate is constant,

$$\begin{aligned} \text{volume} &= (\text{rate})(\text{time}) \\ &= (4 \text{ litres/minute})(3 \text{ minutes}) \\ &= 12 \text{ litres.} \end{aligned}$$

A Constant Rate as Area



The area is

$$(3 \text{ minutes})(4 \text{ litres/minute}) = 12 \text{ litres.}$$

Two Constant Flow Rates

A hose fills a tank at 3 litres/minute for the first minute, then at 5 litres/minute for the next two minutes. How much water enters the tank?

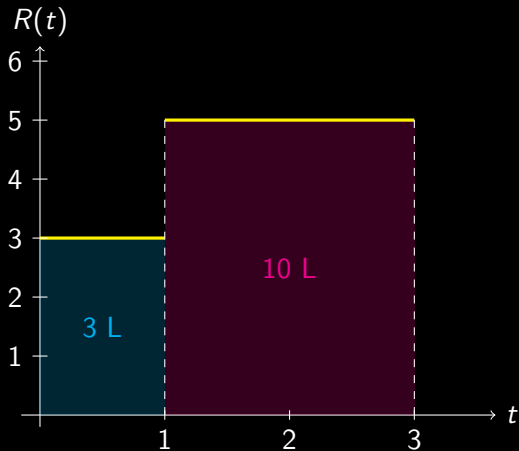
The rate function is

$$R(t) = \begin{cases} 3, & 0 \leq t < 1, \\ 5, & 1 \leq t \leq 3. \end{cases}$$

Calculate the volume on each interval, then add:

$$\begin{aligned} \text{volume} &= (3)(1) + (5)(2) \\ &= 3 + 10 \\ &= 13 \text{ litres.} \end{aligned}$$

Two Rates as Areas



The total area is

$$(1)(3) + (2)(5) = 13 \text{ litres.}$$

Three Constant Flow Rates

A hose fills a tank at 2 litres/minute during the first minute, 5 litres/minute during the second minute, and 4 litres/minute during the third minute. How much water enters the tank?

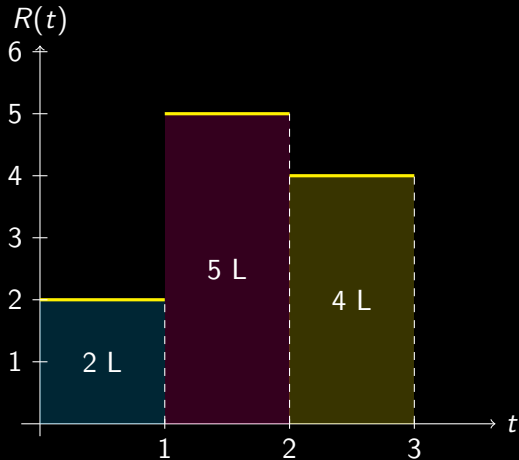
The rate function is

$$R(t) = \begin{cases} 2, & 0 \leq t < 1, \\ 5, & 1 \leq t < 2, \\ 4, & 2 \leq t \leq 3. \end{cases}$$

The volume added is

$$\begin{aligned} \text{volume} &= (2)(1) + (5)(1) + (4)(1) \\ &= 2 + 5 + 4 \\ &= 11 \text{ litres.} \end{aligned}$$

Three Rates as Areas

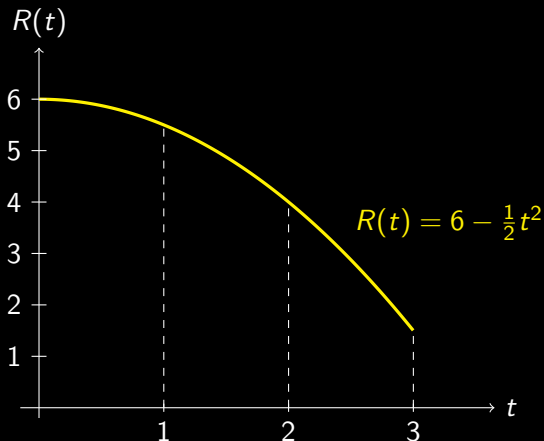


The total area is $2 + 5 + 4 = 11$ litres.

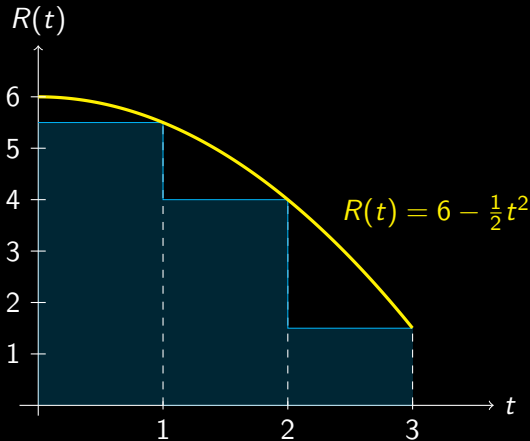
A Continuously Changing Flow Rate

A hose fills a tank for three minutes. Its flow rate decreases continuously according to

$$R(t) = 6 - \frac{1}{2}t^2 \text{ litres/minute}, \quad 0 \leq t \leq 3.$$



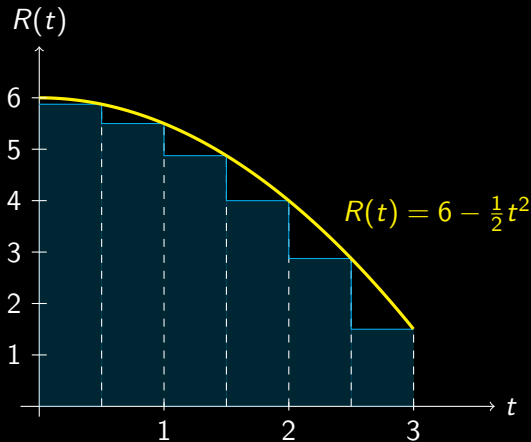
Three Rectangles



Using right endpoints and $\Delta t = 1$,

$$V_3 = 5.5(1) + 4(1) + 1.5(1) = 11 \text{ litres.}$$

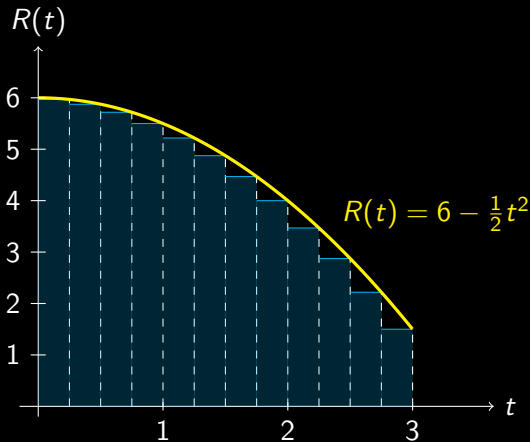
Six Rectangles



With $\Delta t = 0.5$,

$$V_6 = 0.5(5.875 + 5.5 + 4.875 + 4 + 2.875 + 1.5) = 12.3125 \text{ litres.}$$

Twelve Rectangles



Using right endpoints and $\Delta t = 0.25$,

$$V_{12} = 12.921875 \text{ litres.}$$

More Rectangles Improve the Approximation

Divide $0 \leq t \leq 3$ into n equal intervals:

$$\Delta t = \frac{3}{n}, \quad t_k = \frac{3k}{n}.$$

Here, $k = 1, 2, \dots, n$ labels the rectangles from left to right.
The right-endpoint rectangle sum is

$$\begin{aligned} V_n &= \sum_{k=1}^n R(t_k) \Delta t \\ &= \sum_{k=1}^n \left(6 - \frac{1}{2} \left(\frac{3k}{n} \right)^2 \right) \frac{3}{n}. \end{aligned}$$

As n increases, the rectangles become narrower and follow the curve more closely.

Convergence of the Rectangle Sums

$$V_n = \sum_{k=1}^n \left(6 - \frac{1}{2} \left(\frac{3k}{n} \right)^2 \right) \frac{3}{n}$$

Number of rectangles, n	Approximate area, V_n (litres)
3	11
6	12.3125
12	12.921875
24	13.214844
48	13.358398
96	13.429443
192	13.464783
384	13.482407

The Definite Integral

As the number of rectangles tends to infinity,

$$n \rightarrow \infty \quad \text{and} \quad \Delta t = \frac{3}{n} \rightarrow 0.$$

The rectangle sum converges to the definite integral:

$$\begin{aligned} \text{volume} &= \lim_{n \rightarrow \infty} \sum_{k=1}^n R(t_k) \Delta t \\ &= \int_0^3 R(t) dt. \end{aligned}$$

For $R(t) = 6 - \frac{1}{2}t^2$,

$$\begin{aligned} \int_0^3 \left(6 - \frac{1}{2}t^2\right) dt &= \left[6t - \frac{t^3}{6}\right]_0^3 \\ &= 13.5 \text{ litres.} \end{aligned}$$

Volumes of Revolution

Rotating a region about an axis creates a solid.

A thin slice perpendicular to the axis forms a disk. If the disk has radius r and thickness Δs , then

$$\Delta V \approx \pi r^2 \Delta s.$$

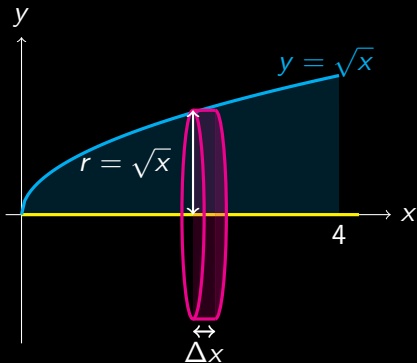
Adding the disks and taking the limit gives

$$V = \lim_{n \rightarrow \infty} \sum_{k=1}^n \pi r_k^2 \Delta s = \pi \int r(s)^2 ds.$$

The direction of the slice determines whether we integrate with respect to x or y .

Volume About the x -Axis

Rotate the region under $y = \sqrt{x}$, for $0 \leq x \leq 4$, about the x -axis.



The highlighted slice forms a thin cylinder of radius $\sqrt{x}$ and thickness Δx .

$$\text{volume of } k\text{th slice} \approx \pi(\sqrt{x_k})^2 \Delta x$$

Volume About the x -Axis

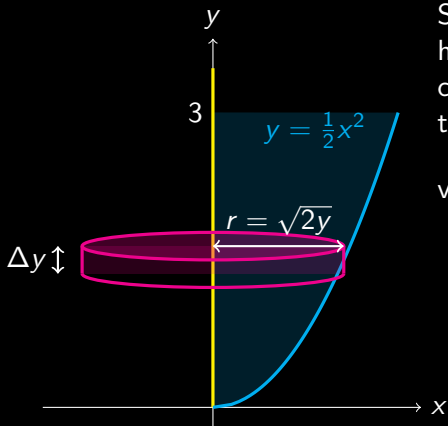
volume of k th slice $\approx \pi(\sqrt{x_k})^2 \Delta x$.

As the number of slices, n , increases the sum of the slice volumes becomes

$$\begin{aligned} V &= \lim_{n \rightarrow \infty} \sum_{k=1}^n \pi(\sqrt{x_k})^2 \Delta x \\ &= \pi \int_0^4 (\sqrt{x})^2 dx \\ &= \pi \int_0^4 x dx \\ &= \pi \left[\frac{x^2}{2} \right]_0^4 \\ &= 8\pi \text{ units}^3. \end{aligned}$$

Volume About the y -Axis

Rotate the region between the y -axis and $y = \frac{1}{2}x^2$, for $0 \leq y \leq 3$, about the y -axis.



Since $x = \sqrt{2y}$, the highlighted slice forms a thin cylinder of radius $\sqrt{2y}$ and thickness Δy .

volume of k th slice $\approx \pi(\sqrt{2y_k})^2 \Delta y$

Volume About the y -Axis

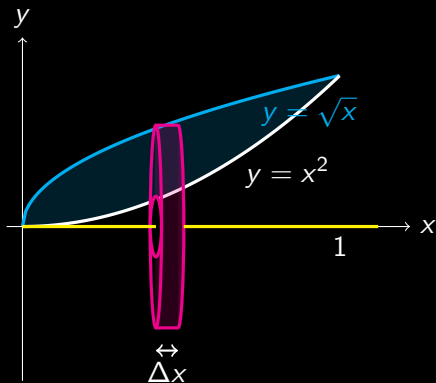
volume of k th slice $\approx \pi(\sqrt{2y_k})^2 \Delta y$.

As the number of slices, n , increases the sum of the slice volumes becomes

$$\begin{aligned} V &= \lim_{n \rightarrow \infty} \sum_{k=1}^n \pi(\sqrt{2y_k})^2 \Delta y \\ &= \pi \int_0^3 (\sqrt{2y})^2 dy \\ &= \pi \int_0^3 2y dy \\ &= \pi [y^2]_0^3 \\ &= 9\pi \text{ units}^3. \end{aligned}$$

Volume Between Two Curves About the x -Axis

Rotate the region between $y = \sqrt{x}$ and $y = x^2$, for $0 \leq x \leq 1$, about the x -axis.



The k th slice has outer radius $R_k = \sqrt{x_k}$ and inner radius $r_k = x_k^2$.

$$\begin{aligned}\text{volume of } k\text{th slice} \\ &\approx \pi R_k^2 \Delta x - \pi r_k^2 \Delta x \\ &= \pi (R_k^2 - r_k^2) \Delta x.\end{aligned}$$

Volume Between Two Curves About the x -Axis

volume of k th slice $\approx \pi ((\sqrt{x_k})^2 - (x_k^2)^2) \Delta x$.

As the number of slices, n , increases, the sum of the slice volumes becomes

$$\begin{aligned} V &= \lim_{n \rightarrow \infty} \sum_{k=1}^n \pi ((\sqrt{x_k})^2 - (x_k^2)^2) \Delta x \\ &= \pi \int_0^1 (x - x^4) dx \\ &= \pi \left[\frac{x^2}{2} - \frac{x^5}{5} \right]_0^1 \\ &= \frac{3\pi}{10} \text{ units}^3. \end{aligned}$$

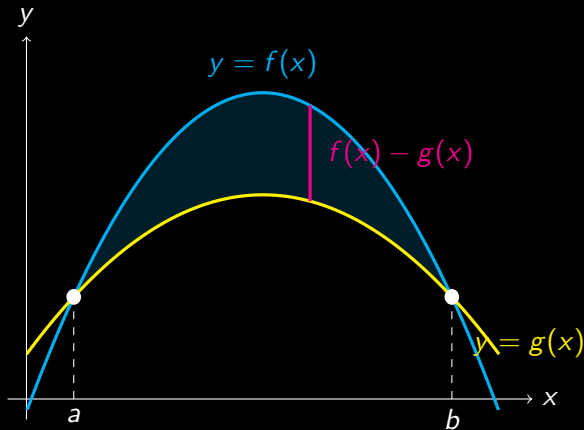
Area Between Two Curves

Suppose $f(x) \geq g(x)$ on $[a, b]$. The area between the curves is

$$A = \int_a^b (f(x) - g(x)) \, dx.$$

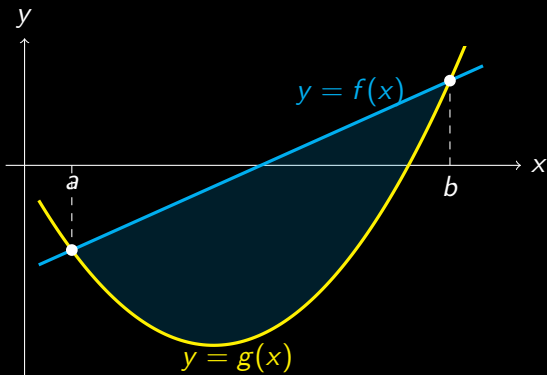
Area is calculated as upper curve minus lower curve.

Area Between Curves: Diagram



$$A = \int_a^b (\text{upper} - \text{lower}) \, dx.$$

Curves That Cross the x -Axis



$$A = \int_a^b (f(x) - g(x)) dx.$$

The formula depends on the vertical distance between the curves, so it still applies when either curve crosses the x -axis.

Area Between Consecutive Intersections

In this lecture, the limits of integration are intersection points.

1. Solve $f(x) = g(x)$.
2. Arrange the intersection values from left to right.
3. On each interval, identify the upper and lower curves.
4. Integrate upper minus lower and add the resulting areas.

A sketch or test value helps identify the upper curve.

Area Between Curves: Example 1

Find the area enclosed by

$$y = 2x \quad \text{and} \quad y = x^2.$$

Find the intersections:

$$\begin{aligned} 2x &= x^2, \\ x(x - 2) &= 0, \quad x = 0, 2. \end{aligned}$$

On $0 \leq x \leq 2$, the line $y = 2x$ lies above the parabola $y = x^2$.

Area Between Curves: Example 1

Continued

Therefore,

$$\begin{aligned} A &= \int_0^2 (2x - x^2) dx \\ &= \left[x^2 - \frac{x^3}{3} \right]_0^2 \\ &= 4 - \frac{8}{3} \\ &= \frac{4}{3}. \end{aligned}$$

The area is $\frac{4}{3}$ square units.

Area Between Curves: Example 2

Find the area enclosed by

$$y = x^2 \quad \text{and} \quad y = 2 - x^2.$$

The curves intersect when

$$\begin{aligned} x^2 &= 2 - x^2, \\ x^2 &= 1, \quad x = -1, 1. \end{aligned}$$

Since $y = 2 - x^2$ is the upper curve,

$$\begin{aligned} A &= \int_{-1}^1 ((2 - x^2) - x^2) dx \\ &= \left[2x - \frac{2x^3}{3} \right]_{-1}^1 = \frac{8}{3}. \end{aligned}$$

Curves That Exchange Positions

Find the area between $y = x$ and $y = x^3$ on $[-1, 1]$.

The curves intersect at $x = -1$, 0 , and 1 . Their order changes at $x = 0$:

$$A = \int_{-1}^0 (x^3 - x) dx + \int_0^1 (x - x^3) dx.$$

$$\begin{aligned} A &= \left[\frac{x^4}{4} - \frac{x^2}{2} \right]_{-1}^0 + \left[\frac{x^2}{2} - \frac{x^4}{4} \right]_0^1 \\ &= \frac{1}{4} + \frac{1}{4} \\ &= \frac{1}{2}. \end{aligned}$$

Improper Integrals

A definite integral is improper when

- ▶ at least one limit of integration is infinite, or
- ▶ the integrand becomes unbounded within the interval.

An improper integral represents a limit. It **converges** if the limit exists and is finite. Otherwise, it **diverges**.

An Infinite Interval

Define

$$\int_a^{\infty} f(x) dx = \lim_{b \rightarrow \infty} \int_a^b f(x) dx.$$

Example:

$$\begin{aligned} \int_0^{\infty} e^{-x} dx &= \lim_{b \rightarrow \infty} \int_0^b e^{-x} dx \\ &= \lim_{b \rightarrow \infty} [-e^{-x}]_0^b \\ &= \lim_{b \rightarrow \infty} (1 - e^{-b}) = 1. \end{aligned}$$

An unbounded region can have finite area.

The p -Integral

For $p > 0$,

$$\int_1^{\infty} \frac{dx}{x^p} = \lim_{b \rightarrow \infty} \int_1^b x^{-p} dx.$$

Value of p	Result
$p > 1$	Converges to $\frac{1}{p-1}$
$0 < p \leq 1$	Diverges

A Divergent Integral

Consider

$$\begin{aligned}\int_1^{\infty} \frac{dx}{x} &= \lim_{b \rightarrow \infty} \int_1^b \frac{dx}{x} \\ &= \lim_{b \rightarrow \infty} [\ln x]_1^b \\ &= \lim_{b \rightarrow \infty} \ln b.\end{aligned}$$

Since $\ln b \rightarrow \infty$, the integral diverges.

The integrand approaches zero, but it does not approach zero quickly enough to produce a finite integral.

An Unbounded Integrand

The integrand $1/\sqrt{x}$ is unbounded at $x = 0$. Define

$$\begin{aligned}\int_0^1 \frac{dx}{\sqrt{x}} &= \lim_{a \rightarrow 0^+} \int_a^1 x^{-1/2} dx \\ &= \lim_{a \rightarrow 0^+} \left[2\sqrt{x} \right]_a^1 \\ &= \lim_{a \rightarrow 0^+} (2 - 2\sqrt{a}) = 2.\end{aligned}$$

The integral converges even though the integrand is unbounded at one endpoint.

An Interior Singularity

If f is unbounded at $x = c$ inside $[a, b]$, split the integral:

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx.$$

Evaluate each part using its own one-sided limit:

$$\begin{aligned}\int_a^c f(x) dx &= \lim_{u \rightarrow c^-} \int_a^u f(x) dx, \\ \int_c^b f(x) dx &= \lim_{v \rightarrow c^+} \int_v^b f(x) dx.\end{aligned}$$

The original integral converges only when both limits converge.

For $\int_{-1}^1 \frac{dx}{x^2}$, split the integral at $x = 0$.

An Interior Singularity: Example

Evaluate the first one-sided integral:

$$\begin{aligned}\int_{-1}^0 \frac{dx}{x^2} &= \lim_{u \rightarrow 0^-} \int_{-1}^u x^{-2} dx \\ &= \lim_{u \rightarrow 0^-} \left[-\frac{1}{x} \right]_{-1}^u \\ &= \lim_{u \rightarrow 0^-} \left(-\frac{1}{u} - 1 \right).\end{aligned}$$

Since this one-sided integral diverges,

$$\int_{-1}^1 \frac{dx}{x^2}$$

diverges.

Week 10 Summary

- ▶ Accumulated volume is the area under a flow-rate graph.
- ▶ Area between curves uses upper minus lower between intersection points.
- ▶ Improper integrals are defined using limits and may converge or diverge.
- ▶ Disk slices calculate volumes of revolution about either coordinate axis.
- ▶ For a region between two curves, subtract the inner disk from the outer disk.